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SCCAD-3N and OpenROAD Enablement for Open-Source Advanced-Node Design

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Open-source 3nm PDK

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USC-3N-2D (3nm PDK for 2D ICs)

USC-3N-2D is an open-source 3nm Process Design Kit (PDK) for 2D ICs developed by the [Southern California Computer-Aided Design \(SCCAD\) Lab](#) at the University of Southern California (USC). This PDK is intended for academic research and educational use, and enables exploration of advanced-node physical design, device–interconnect interactions, and design–technology co-optimization (DTCO). Our key features include:

<https://github.com/SCCAD-LAB/USC-3N-2D>

- Dual-side BEOL support – Enables backside routing for power, clock, and signals. Our standard cells for backside power delivery networks (BS-PDN) integrate buried power rails (BPR) to improve routing efficiency and power integrity.
- Broad tool compatibility – Compatible with [OpenROAD](#), [Cadence](#), and [Synopsys](#) physical design flows. Reference collateral includes [sample layouts & PPA data](#), netlists, flow scripts, and DEF files for 6 designs.



Goals and Features

- **Goals**
 - Open-source 3nm PDK for research and education.
 - Enable advanced-node physical design and DTCO research.
 - Accelerate exploration of new design methodologies.
- **Features**
 - Customizable: Modify devices, interconnects, cells, and P&R flows.
 - Dual-side BEOL: Supports backside routing with BPR-based standard cells.
 - Tool-compatible: Works with OpenROAD, Cadence, and Synopsys; includes reference designs and flow collateral.



Methodology

- Device modeling
 - TCAD-calibrated nanosheet FETs translated to BSIM-CMG compact models.
- Interconnect modeling
 - IRDS-scaled BEOL with literature-based RC values and extracted parasitics.
- DRC/PEX
 - Predictive rule decks based on GT3/ASAP7
 - Extended for 3 nm GAAFET, BPR, and backside PDN.



Use Cases

- DTCO research
 - Evaluate advanced devices, interconnects, and power delivery.
- EDA development
 - Benchmark Cadence, Synopsys, and OpenROAD at a 3 nm–like node.
- Architecture exploration
 - Enable end-to-end digital implementation and cross-node studies with ASAP7.



Ways to Customize the PDK

- Open & modular
 - Edit device models, interconnects, standard cells, and PnR flows.
- Full-stack exploration
 - Evaluate changes with the provided end-to-end flow and sample designs.
- Flexible experimentation
 - Transparent, manually editable framework for research.



Room for Improvement

- Predictive models
 - Device/interconnect models are not silicon-calibrated.
- Limited collateral
 - Small standard cell library and simplified signoff infrastructure.
- Future work
 - Improve calibration, expand libraries, and add variation/manufacturing effects (community contributions welcome).



Example PnR Runs

Tool	Design	README	Data Files
Cadence	ECG	README	ECG
	OpenPiton	README	OpenPiton
Synopsys	ECG	README	ECG
	OpenPiton	README	OpenPiton
OpenROAD	ECG	README	ECG
	OpenPiton	README	OpenPiton





Upcoming Releases

- USC-3N-3D:
 - 3nm PDK, ADK, and scripts for 3D IC design and simulation
- USC-3N-25D:
 - 3nm PDK, ADK, and scripts for 2.5D IC design and simulation

